

LLM-Based Volunteer Coordination for Crowd Evacuation: A Comparison with a Game-Theoretic Model

110652012 施品光

Introduction

Crowd evacuation simulation is an important research topic for improving public safety during emergencies such as fires and earthquakes. Existing evacuation models commonly employ Cellular Automata and Floor Field methods to simulate pedestrian movement and evaluate evacuation efficiency.

In this work, we investigate whether a Large Language Model (LLM) can improve volunteer coordination in crowd evacuation. We replace the original game-theoretic volunteer decision module with an LLM-based reasoning framework while preserving the remaining simulation components for a fair comparison. Experimental results demonstrate improved evacuation efficiency together with interpretable volunteer decisions.

The main contributions of this work are summarized as follows:

1. We replace the original game-theoretic volunteer decision model with an LLM-based reasoning framework.
2. We preserve the original evacuation framework to enable a fair comparison.
3. We improve evacuation performance while providing interpretable volunteer decisions.

Background

II. Background

The proposed work is based on the crowd evacuation framework introduced by Lin and Wong [1]. The simulation environment consists of multiple exits, some of

which may be blocked by obstacles. Pedestrians move toward available exits using Cellular Automata (CA) and Floor Field guidance. Unlike conventional evacuation models, the framework allows pedestrians to voluntarily remove obstacles, enabling blocked exits to become available again and improving evacuation efficiency.

To determine volunteer behavior, the original model adopts the Volunteer's Dilemma, a game-theoretic approach that estimates the probability of an agent becoming a volunteer based on predefined mathematical equations. Potential volunteers are first identified, after which a volunteer is assigned to remove each obstacle. Although this approach effectively models volunteer behavior, its decision process mainly relies on predefined rules and limited local information. This limitation motivates the use of LLM-based reasoning for volunteer coordination, which is introduced in the following section.

Purpose Method

Original volunteer coordination

The original model determines volunteer behavior using the Volunteer's Dilemma. During each selection step, pedestrians first identify nearby obstacles and visible agents. A game-theoretic decision model is then applied to estimate the willingness of each pedestrian to become a volunteer.

Potential volunteers are grouped according to the obstacle they can assist. For each obstacle, the nearest volunteer is selected to remove the blockage, while all remaining pedestrians continue their original evacuation behavior.

After the volunteer assignment is completed, selected volunteers move toward the assigned obstacle and remove it. Once the obstacle is cleared, the corresponding exit becomes available again, allowing subsequent pedestrians to evacuate more efficiently. This volunteer coordination mechanism serves as the baseline for comparison in this study.

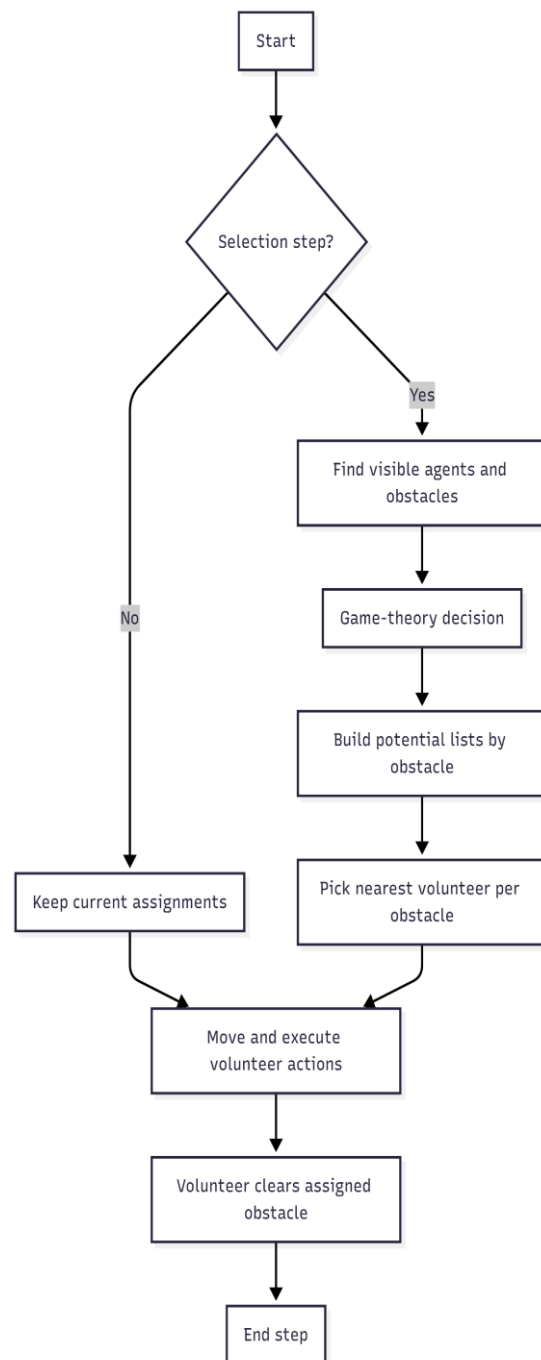


Fig. 1

$P_{vol} = 1 - (1 - e^{-\omega b})^{1/x} \cdot \beta^{1/(x-1)}$ $\beta = \text{clamp}\left(\frac{C_v}{C_f}, 10^{-4}, 1\right)$ $x = \max(2, N_{near})$ $Volunteer = \mathbf{1}[U(0, 1) < P_{vol}]$	N_{near}: Number of nearby pedestrians B: Obstacle blockage ratio ω: Unwillingness coefficient C_v: Volunteer cost C_f: Failure cost B: Cost ratio C_v/C_f
--	--

Fig. 2

Equation estimates the probability that a pedestrian becomes a volunteer. The decision is influenced by the local evacuation context, including the number of nearby pedestrians, the obstacle blockage ratio, and the relative cost of volunteering. A higher blockage ratio generally increases the willingness to volunteer, while a higher volunteer cost decreases the volunteer probability. Finally, a Bernoulli sampling process is performed to determine whether the pedestrian becomes a volunteer.

After volunteer probabilities are calculated, potential volunteers are grouped according to each obstacle. The nearest volunteer is then assigned to remove the obstacle while the remaining pedestrians continue evacuation.

Proposed LLM Volunteer Coordination

Instead of using a predefined mathematical model, the proposed approach employs an LLM to determine volunteer assignments. The LLM reasons over the current evacuation context and generates volunteer coordination decisions.

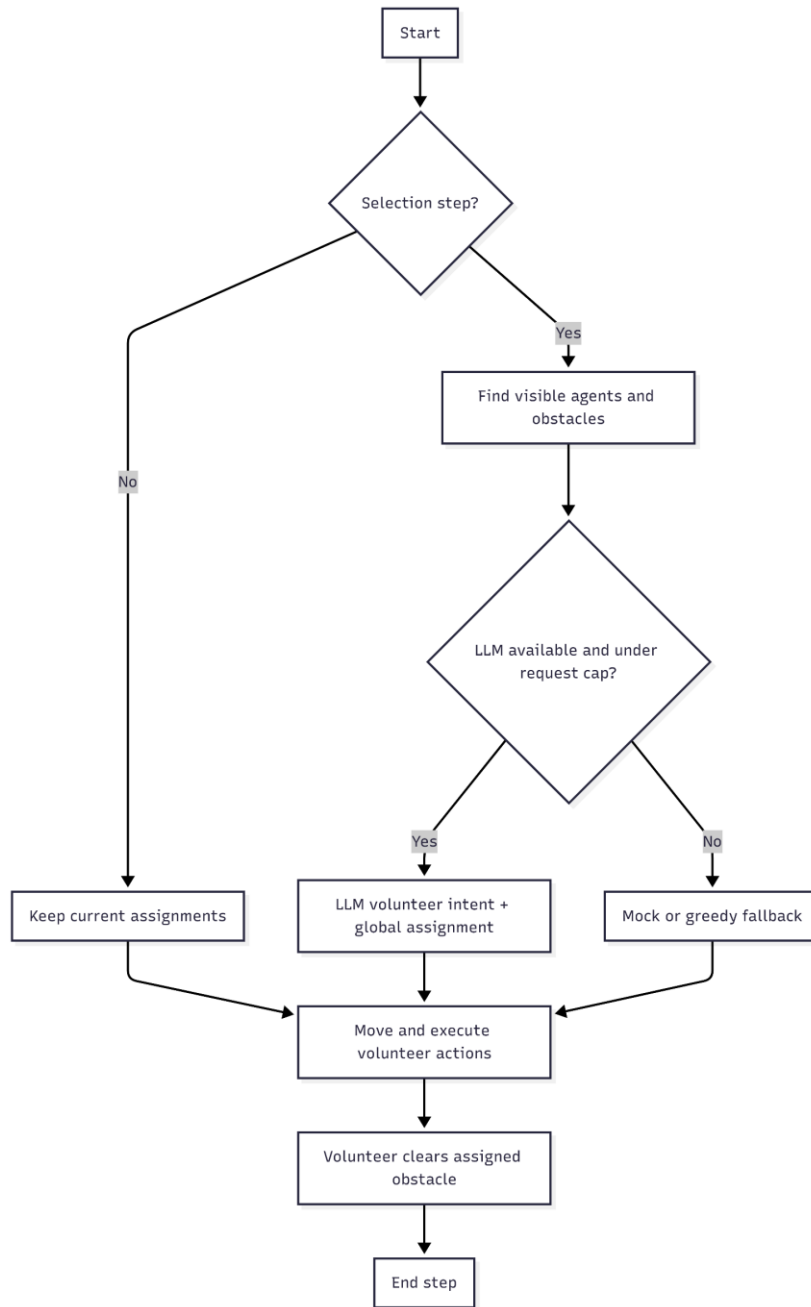


Fig. 3

As illustrated in Fig. 3, the proposed framework first collects the current evacuation context, including visible obstacles and nearby pedestrians. If a volunteer selection step is triggered, the collected information is provided to the LLM, which generates volunteer assignments. Otherwise, existing assignments are preserved.

Input	Description
Visible Obstacles	Obstacles within observation range
Nearby Pedestrians	Pedestrians around each obstacle
Available Volunteers	Candidate volunteers
Distance	Distance between volunteer and obstacle
Obstacle Impact	Estimated influence on evacuation

Unlike the original mathematical model, the LLM simultaneously considers multiple contextual factors when making decisions. Table 2 summarizes the information provided to the LLM during each volunteer selection step.

Based on the provided context, the LLM determines volunteer assignments for the current evacuation state. Compared with the original obstacle-local strategy, the proposed method performs global volunteer coordination by considering multiple obstacles and candidate volunteers simultaneously.

In addition to volunteer assignments, the LLM also generates natural-language explanations describing the reasoning behind each decision. These explanations improve the interpretability of the proposed framework and facilitate analysis of volunteer behavior.

Framework

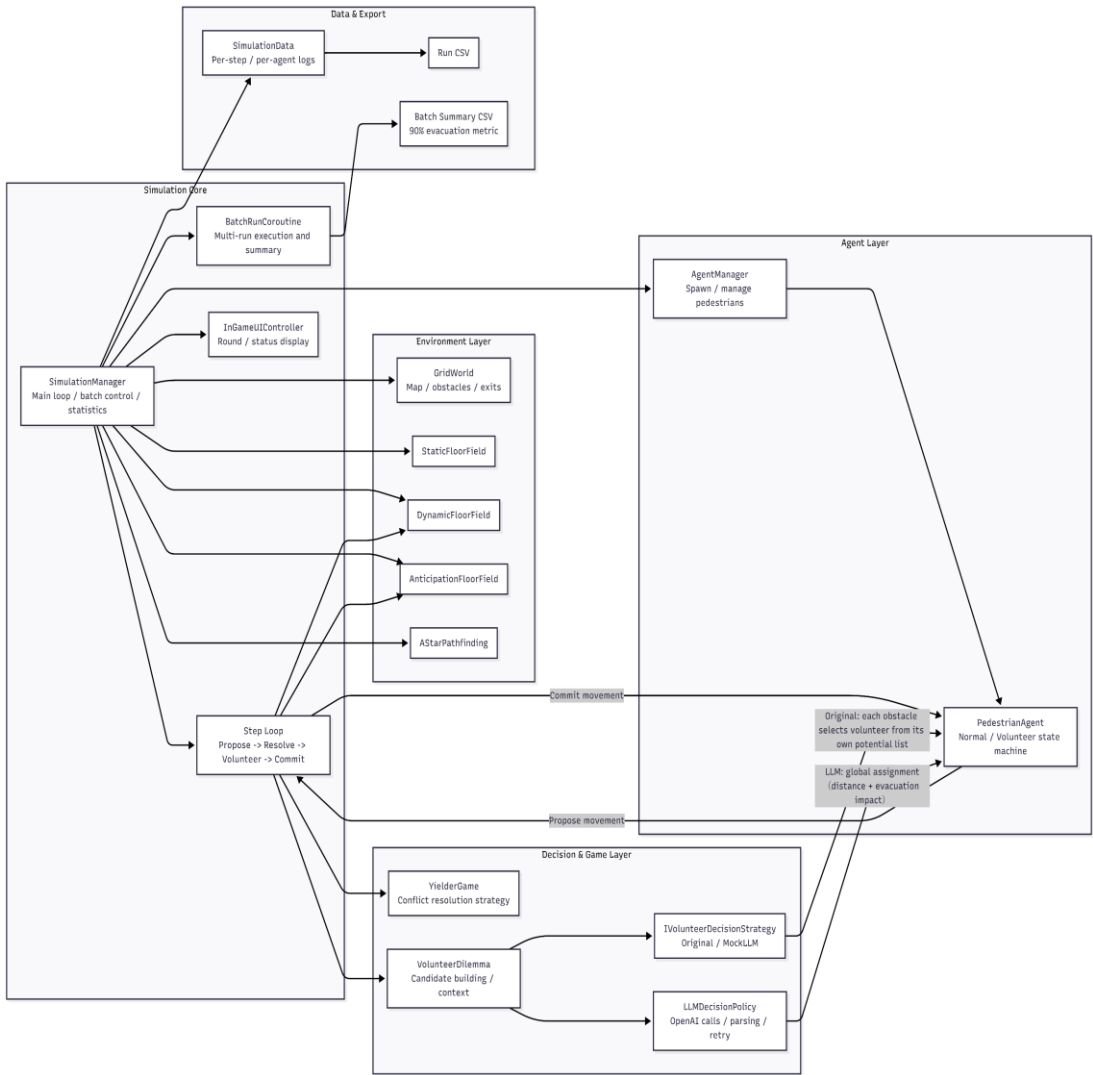


Fig. 4

Fig 4 illustrates the overall architecture of the proposed evacuation framework. The simulation consists of five major components: the Simulation Core, Environment Layer, Agent Layer, Decision Layer, and Data Export module.

Layer	Function
Simulation Core	Controls the simulation loop
Environment Layer	Manages map and floor fields
Agent Layer	Controls pedestrian behaviors
Decision Layer	Performs volunteer coordination
Data Export	Records simulation results

The Simulation Core manages the overall simulation loop. The Environment Layer maintains map information and floor-field guidance. The Agent Layer controls pedestrian movement, while the Decision Layer performs volunteer coordination. Finally, the Data Export module records simulation statistics for evaluation.

Only the volunteer decision module is modified in this work.

Compared with the original framework, all simulation components remain unchanged except for the volunteer decision module. This design minimizes implementation bias and enables a fair comparison between the original and proposed approaches.

Explainable Volunteer Reasoning

Unlike the original mathematical model, the proposed approach not only generates volunteer assignments but also provides natural-language explanations describing the reasoning behind each decision. This additional information improves the interpretability of the volunteer coordination process

Step	AgentID	Decision	Explanation
1	-1239688	selfish	The agent is an evacuee with a relatively short distance to the exit and is currently not positioned near the target obstacle. Given the high number of nearby obstacles (20) and the presence of other evacuees near the target c
1	-1239702	volunteer	The agent is close to the target obstacle and there are several evacuees nearby who may benefit from the removal of the obstacle. With a significant estimated evacuation impact of 0.64 and 6 evacuees near the target obsta
1	-1239716	volunteer	The agent is positioned close to the exit and has a reasonable chance of helping to remove the nearby obstacle. With a high estimated evacuation impact of 0.68 and multiple evacuees near the target obstacle, volunteering t
3	-1239716	volunteer	assigned volunteer cleared obstacle
4	-1239702	volunteer	assigned volunteer cleared obstacle
10	-1239688	volunteer	The agent is positioned close to the target obstacle and there are many evacuees nearby. By volunteering to remove the obstacle, the agent can significantly impact the overall evacuation process, especially since the estima
10	-1239730	selfish	The agent is an evacuee located 6 units from the exit and has a considerable distance of 9 units to the target obstacle. With 5 evacuees near the target obstacle and only 1 current volunteer, the agent's involvement would no
10	-1239744	volunteer	The agent is situated close to the exit (3 units away) and there are nearby obstacles that are impacting the evacuation process. By volunteering to remove the target obstacle at (25, 12), which is affecting 8 other evacuees, t
10	-1239688	volunteer	assigned volunteer cleared obstacle
20	-1239856	volunteer	The agent is in a position where removing the nearby target obstacle (1, 15) would significantly benefit the evacuation effort, as it is only 1 unit away from the exit and there are 7 evacuees near it. The estimated evacuatio
20	-1239884	volunteer	The agent is only 2 units away from the exit and has a high estimated evacuation impact of 0.63. By volunteering to remove the nearby obstacle, the agent can significantly help themselves and others evacuate more effici
20	-1239908	selfish	The agent is only 3 units away from the exit and has a high estimated evacuation impact of 0.74. Given that there are already 2 current volunteers and 6 evacuees near the target obstacle, the agent should prioritize their on
21	-1239856	volunteer	assigned volunteer cleared obstacle
21	-1239884	volunteer	assigned volunteer cleared obstacle
30	-1240034	volunteer	The agent is in a critical position with a distance of 3 to the exit and can significantly impact the evacuation by removing a nearby obstacle. With 2 evacuees near the target obstacle and an estimated evacuation impact of 0
30	-1240178	volunteer	The agent is very close to the exit (distance of 1) and has a chance to assist in removing an obstacle (target obstacle at (1, 16)) that could potentially help other evacuees. The estimated evacuation impact of 0.65 indicates a
31	-1240178	volunteer	assigned volunteer cleared obstacle

Fig 5 (Example of LLM-generated volunteer reasoning)

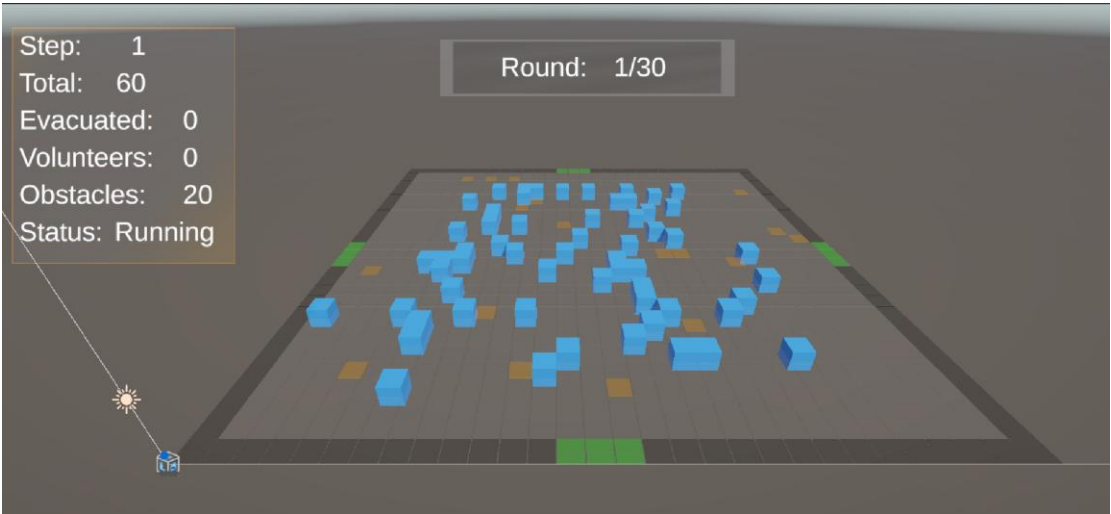
The generated reasoning allows users to understand why a particular volunteer is selected. Instead of only producing a final assignment, the LLM explicitly describes the contextual factors considered during decision making, such as obstacle impact, volunteer distance, and surrounding pedestrian density.

Although the reasoning itself does not directly affect simulation performance, it provides additional transparency that is unavailable in the original model. Such explainable outputs facilitate result analysis and may support future human-AI collaborative evacuation planning.

Experimental Setup

Simulation Environment

Parameter	Value
Environment	Grid-based
Exits	4
Obstacles	Randomly generated
Obstacles Counts	20, 40, 60, 80
Agent Counts	60, 90, 120, 150
Runs	30



Experiments were conducted in a grid-based evacuation environment with four exits and randomly generated obstacles. Four crowd densities (60, 90, 120, and 150 agents) were evaluated. Each scenario was repeated 30 times to reduce randomness, and volunteer selection was performed every six simulation steps.

Evaluation Metric

Performance was evaluated using the T90 evacuation time, which represents the number of simulation steps required for 90% of pedestrians to successfully evacuate. A lower T90 value indicates better evacuation efficiency.

Experimental Results

Evacuation Performance

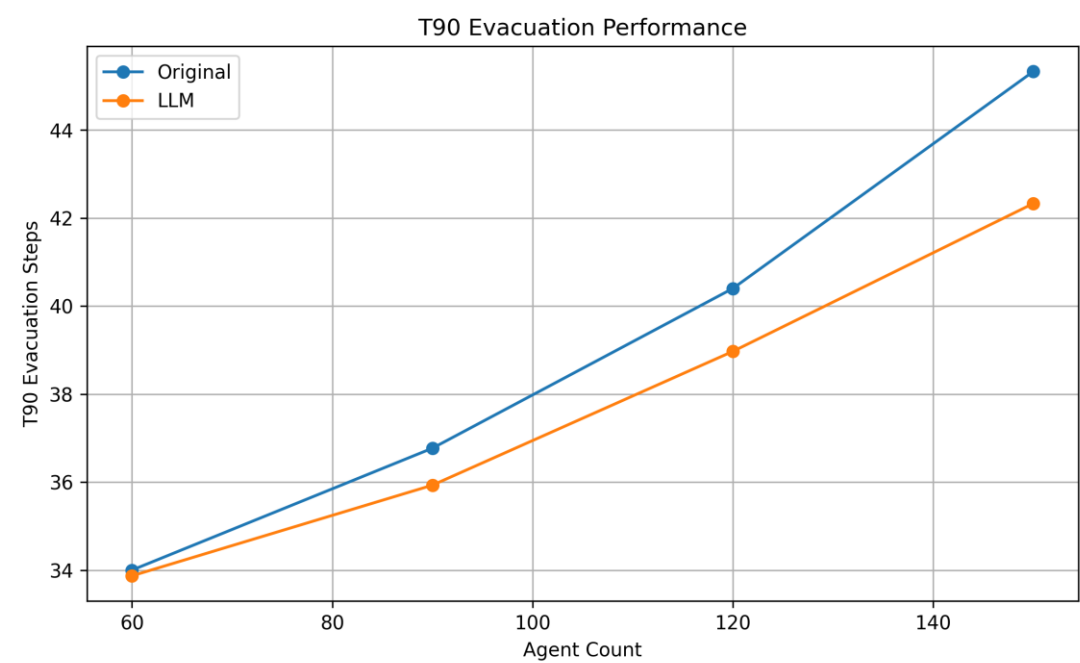


Fig 7.

Fig 7 compares the T90 evacuation time of the original model and the proposed LLM-based approach under different crowd densities. Lower T90 values indicate better evacuation performance.

The proposed method consistently achieves lower T90 values than the original model across all scenarios. The performance gap becomes more significant as the number of pedestrians increases, indicating that the LLM-based coordination is more effective in dense evacuation environments.

Performance Improvement

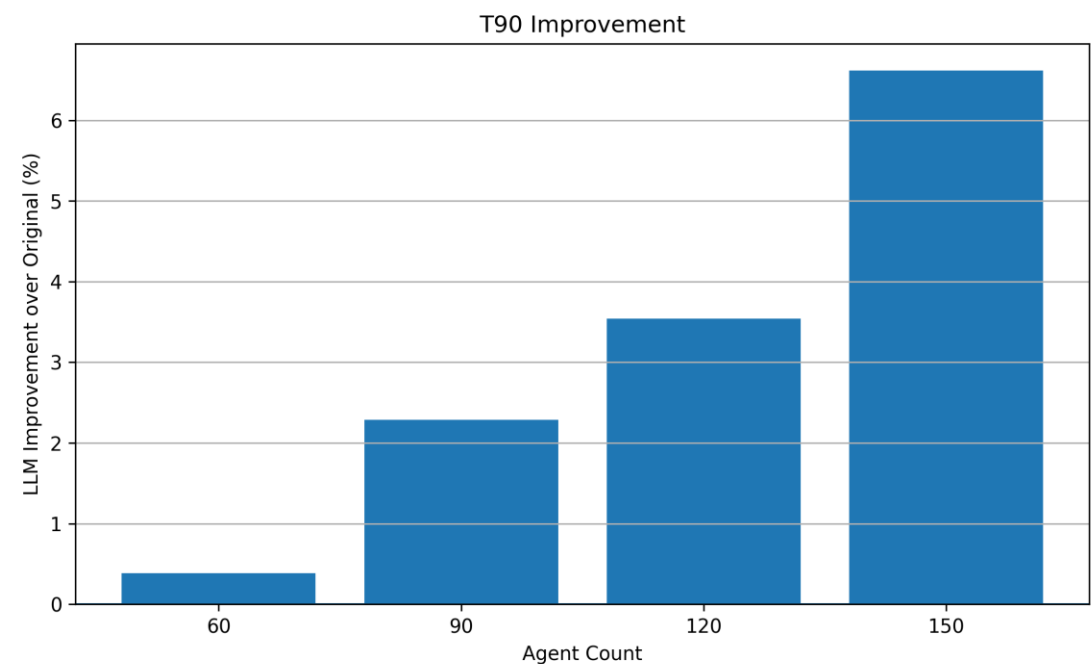


Fig 8

Fig 8. Relative improvement of the proposed method over the original model.

The improvement is relatively small in low-density scenarios because volunteer conflicts are limited. As crowd density increases, the advantage of global volunteer coordination becomes more apparent, resulting in larger performance gains.

Qualitative Analysis

In addition to improving evacuation performance, the proposed framework provides interpretable volunteer decisions. Figure 9 presents an example of the reasoning generated by the LLM during volunteer coordination.

Unlike the original mathematical model, which only outputs the final volunteer assignment, the LLM explicitly explains why a volunteer is selected. Typical reasoning includes obstacle impact, volunteer proximity, and the number of affected pedestrians.

Although these explanations do not directly affect evacuation performance, they

provide additional transparency that is unavailable in the original model. Such interpretable outputs facilitate behavior analysis and may support future human-AI collaborative evacuation planning.

Discussion

Advantages

Global Coordination

Unlike the original model, which assigns volunteers independently for each obstacle, the proposed framework considers all obstacles and candidate volunteers simultaneously. This global coordination helps reduce redundant assignments and improves overall evacuation efficiency.

Context Awareness

The LLM utilizes richer contextual information, including obstacle impact, volunteer distance, and nearby pedestrian density. This enables more adaptive decisions under different evacuation scenarios.

Explainability

In addition to volunteer assignments, the proposed framework provides natural-language reasoning that explains the decision process. Such explanations improve transparency and facilitate result analysis.

Limitations

API Cost

Inference Time

Prompt Dependency

The quality of volunteer assignments depends on the prompt design and the capability of the selected LLM.

Conclusion

This work presented an LLM-based volunteer coordination framework for crowd evacuation simulation. By replacing the original game-theoretic volunteer decision module while preserving the remaining simulation framework, a fair comparison between the original and proposed approaches was achieved. Experimental results demonstrated that the proposed method consistently reduced the T90 evacuation time under different crowd densities. Furthermore, the LLM provided interpretable decision reasoning, improving the transparency of volunteer coordination. These results suggest that LLM-based reasoning is a promising approach for enhancing volunteer decision-making in crowd evacuation simulations.

Future work

Several directions can be explored in future research. First, more complex evacuation scenarios, such as dynamic obstacles and changing environments, could be considered to evaluate the robustness of the proposed framework. Second, different LLMs or locally deployed language models may be investigated to reduce inference cost and improve response time. Finally, integrating real-world evacuation data and human behavioral studies could further validate the effectiveness and practicality of the proposed volunteer coordination framework.

Reference

[1] Lin, G. W., & Wong, S. K. (2018). Evacuation simulation with consideration of obstacle removal and using game theory. *Physical Review E*, 97(6), 062303.